

# AGUS SALIM

## AI Automation & Agentic Systems Specialist

OPEN TO REMOTE

📍 Central Java, Indonesia (UTC+7)

📄 GitHub: [github.com/genjusniper](https://github.com/genjusniper)

📞 Contact: Available via Upwork & GitHub

⚡ Focus: Agentic Workflows • Node.js • LLM APIs

### PROFESSIONAL PROFILE

**Hands-on AI Automation & Agentic Systems Developer** specializing in building autonomous workflows, custom API integrations, and robust **Human-in-the-Loop (HITL)** business automation. Experienced in architecting end-to-end multi-agent pipelines that bridge raw unstructured web data (Google Maps scraping) with contextual LLM reasoning, strict quality guardrails, and secure messaging gateways (WhatsApp / Baileys). Passionate about pragmatic software development: prioritizing zero-hallucination reliability, domain-accurate vernacular, and resilient production deployments on headless Linux/Termux environments.

### CORE COMPETENCIES & TECHNOLOGY STACK

#### 📁 AGENTIC AI & LLM SYSTEMS

Autonomous Agents, Multi-stage Reasoning, Tool/Function Calling, Context-Aware Pitch Engines, Anti-Hallucination Guardrails, Human-in-the-Loop Approval Loops, Decision Traces, Evidence Scoring.

#### ⚙️ BACKEND & ARCHITECTURE

JavaScript (ESM/CJS), Node.js, Event-Driven Systems (EventBus / PubSub), Asynchronous Queue Workers, Rate Limiting & Debouncers, REST APIs, JSON Schema Validation, Structured Ledgers.

#### 🔍 DATA INTAKE & SCRAPING

Google Maps Scraping (SerpApi), Geographic Targeting, Public Contact Extraction, Business Category Inference, Product Taxonomy Normalization, Evidence Confidence Scoring.

#### 🚀 DEVOPS & INFRASTRUCTURE

Headless Linux & Android Termux Deployments, PM2 Process Clustering & 24/7 Daemonizing, Remote Automation via SSH (ssh2/node-ssh), Git Version Control, Baileys WebSocket Protocol.

### FEATURED FLAGSHIP PROJECT

#### Autonomous B2B Sales Intelligence & Lead Discovery Platform [GitHub: wa-bot-super](#) (Branch: [phase-k-admin-ui](#))

PRODUCTION DEPLOYED

An end-to-end agentic sales intelligence platform engineered to discover local food and beverage businesses via Google Maps, evaluate multi-product supply-demand fit, generate hyper-localized context-aware outreach copy, and enforce cryptographically guarded Human-in-the-Loop approval before dispatching messages.

- ▶ **Automated Lead Scraping & Tool Calling:** Integrates Google Maps API (SerpApi) with heuristic query builders, parsing verified business names, categories, and direct contacts while filtering duplicates.
- ▶ **Domain Demand Inference Engine (BusinessDemandEngine):** Maps business categories (Warteg, Padang, Gudeg, Penyetan, Gorengan) to ingredient requirements (Gori Cacah, Bawang Kupas, Lombok, Daun Singkong) with prioritized confidence scoring.
- ▶ **Dynamic Context-Aware LLM Pitching (ContextAwarePitchEngine):** Crafts custom first-contact sales pitches with multi-product basket recommendations. Strictly removes overpromising buzzwords ("fresh/segar/muda") in favor of authentic wholesale market value propositions.
- ▶ **Human-in-the-Loop Approval Firewall (SalesGuard & ApprovalEngine):** Protects brand reputation by holding all drafts in an administrative quarantine. Dispatches require explicit approval commands (/approve <lead-id>) generating single-use cryptographic security tokens.
- ▶ **Direct WhatsApp Admin UI:** Enables business owners to command discovery pipelines (/leads [location]) and approve dispatches directly within their personal WhatsApp chats.

**Key Engineering Result:** Operates fully headless 24/7 on low-power mobile Linux (Termux PM2), performing real-time discovery in under 3 seconds per lead with 0 unauthorized or hallucinated outbound transmissions.

Node.js (ESM)

Google Gemini API

SerpApi (Google Maps)

Baileys WebSocket

Event-Driven Architecture

Human-in-the-Loop (HITL)

PM2 Daemon

Termux / Linux

## SECOND PRODUCTION PROJECT

### Enterprise WhatsApp Co-Pilot & Resilient Gateway OS

Full-Stack Async Architecture

#### PRODUCTION V15

A high-throughput, multi-stage WhatsApp gateway designed to handle hundreds of daily incoming messages with stateful conversation management, media vision processing, and automated failover.

- ▶ **Finite State Machine (ConversationFSM):** Implements strict state transitions (IDLE → THINKING → RESPONDING) to eliminate race conditions, double-replies, and message loops.
- ▶ **Chat Burst Aggregation (ChatBurstAggregator):** Buffers rapid sequential messages from users within a 2.5-second sliding window, merging text, audio, and images into a single cohesive job context.
- ▶ **Owner Presence Engine & Anti-Interference:** Dynamically detects when the human owner manually types in a chat, immediately stepping down the AI agent to prevent conflicting automated responses.
- ▶ **Multimodal Ingestion & Vision:** Automatically downloads, base64-encodes, and extracts context from photos, receipts, and voice notes for LLM multimodal processing.
- ▶ **Maintenance & Storage Auto-Pruning:** Built-in 6-hour cron service pruning stale trace logs, temporary media cache, and expired SQLite records to keep device storage lightweight.

Baileys WebSocket

Finite State Machine (FSM)

Async JobQueue

Multimodal Media Ingestion

Storage Pruner

WebCockpit (Express)

## ARCHITECTURAL PHILOSOPHY & SAFETY INVARIANTS

Every automation system I develop is built upon strict architectural principles that ensure business safety, eliminate hallucinations, and guarantee predictable production behavior:

#### 🔒 Strict Human-in-the-Loop

AI agents propose drafts; humans approve. No external emails or WhatsApp messages are ever sent without an authenticated cryptographic token issued by the operator.

#### 🌐 Grounded Truth & Evidence

Pitches and actions are derived solely from verifiable evidence (Google Maps categories, public menus). Claims of non-existent items or fake customer demand are rejected at the firewall level.

#### ⚡ Resilient & Low Overhead

Architected to run reliably on resource-constrained hardware (e.g. Android Termux ARM64) without requiring expensive monthly cloud server bills.

## WHY WORK WITH ME? (FREELANCE & REMOTE VALUE)

- ▶ **Relentless Execution & Ownership:** I don't stop at theoretical designs. I write code, test edge cases, deploy on real infrastructure, and refine until the solution solves the actual business bottleneck.
- ▶ **Hyper-Fast Communication & Transparent Logs:** You will always have visibility into project status through clean commit histories, pull requests, structured log outputs, and verifiable demos.
- ▶ **Pragmatic Commercial Mindset:** I understand that technology is an instrument to increase sales and save human labor—not just an academic exercise. I tailor solutions to real user psychology and local market language.

### Ready to Automate Your Business Workflows?

Available for freelance contracts, custom AI automation builds, lead scraping pipelines, and WhatsApp integration projects. Reach out on Upwork or explore my public code repositories at [github.com/genjusniper](https://github.com/genjusniper).